

Tuesday Warm-up: Potentials

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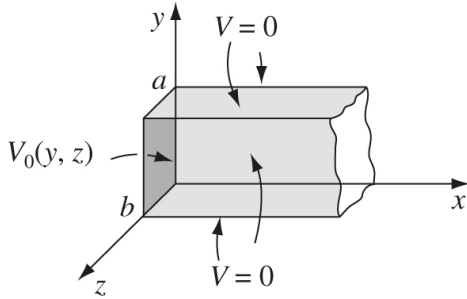


Figure 1: A waveguide with grounded sides and a potential $V_0(y, z)$ on one side.

$x < 2\pi$. Draw a graph of the *spectrum*. That is, draw axes labeled *amplitude* versus *frequency*, and make a mark with appropriate height at every frequency in the series.

1 Memory Bank

- **Fourier series** for a function $f(x)$:

$$f(x) = \frac{A_0}{2} + \sum_{n=1}^{\infty} (A_n \cos nx + B_n \sin nx) \quad (1)$$

$$A_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos(nx) dx \quad (n = 0, 1, 2, \dots) \quad (2)$$

$$B_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin(nx) dx \quad (n = 1, 2, \dots) \quad (3)$$

- **Trigonometric identities:**

$$\cos \theta \cos \phi = \frac{1}{2} (\cos(\theta - \phi) + \cos(\theta + \phi)) \quad (4)$$

$$\sin \theta \sin \phi = \frac{1}{2} (\cos(\theta - \phi) - \cos(\theta + \phi)) \quad (5)$$

$$\sin \theta \cos \phi = \frac{1}{2} (\sin(\theta + \phi) + \sin(\theta - \phi)) \quad (6)$$

$$\cos \theta \sin \phi = \frac{1}{2} (\sin(\theta + \phi) - \sin(\theta - \phi)) \quad (7)$$

2 Separation of Variables

1. **Fourier Series.** Let $\delta_{nn'} = 1$, if $n = n'$, and 0 if $n \neq n'$. (a) Show that the sine functions are *orthogonal*, using an identity in the Memory Bank. That is, prove that

$$\int_0^a \sin(n\pi x/a) \sin(n'\pi x/a) dx = \frac{a}{2} \delta_{nn'} \quad (8)$$

(b) Create the Fourier series for a square wave. That is, let $f(x) = 1$ if $0 \leq x \leq \pi$, and $f(x) = 0$ if $\pi \leq$

2. An infinitely long rectangular metal pipe (sides a and b) is grounded, but one end, at $x = 0$, is maintained at a specified potential $V_0(y, z)$, as indicated in Fig. 1. Find the potential inside the pipe.

- Write the six boundary conditions as equations:

- Write the solution as $v(x, y, z) = X(x)Y(y)Z(z)$, and separate the variables.

- Set the ordinary X , Y , and Z differential equations equal to constants that sum to zero.

- Show that the specific solution is proportional to a decaying exponential in x , and the product of sine functions in y and z .